Summarization: Explicitly Encouraging Low Fractional Dimensional Trajectories via Reinforcement Learning

Summarized by Xiaonan (Nice) Wang *
Summarization†Generated based on Paper from Gillen and Byl. (2021)
Topic: Legged Locomotion, Fractal Geometry, Reinforcement Learning

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Abstract

This document summarizes the core contributions and methodology of the paper "Explicitly Encouraging Low Fractional Dimensional Trajectories via Reinforcement Learning, Gillen and Byl [2021]", focusing on its' main ideas and the core blocks.

1 Overview: Core Questions and Answers

(1) What is the problem?

- In *Legged Locomotion*: Learning an intelligent gait that possesses both **low-dimensional stability** and **high adaptability** within a **stochastic real-world** environment.
  - via **Fractional Dimensionality Reduction**,
  - facilitating **Formal Verification**,
  - and being **robust** to *Noise & Disturbances* (**stability** or **robustness**).

(2) Why need to solve this problem?

- **Robustness and Stability**: Guarantee the system is safe in all the regions of state space during its life.
- Breaking the **curse of dimensionality**: Dealing with scenarios with *high (topological) dimensional state space*.
- Addressing the **Unexplainable** and **Unverifiable** issues of **Model-Free approaches**.

(3) How is it different from prev.?

- When Box Meshing: Storing mesh points in a **Hash Table**.
  - Time Complexity: **Reducing** from $o(n^2)$ to $o(n)$;
  - **Data Compression**: $cnt(keys) \ll n$, points belong to same box will share the same key.
- When Post Processing: "**Divided** by fractional dimension" **rather** than "*Subtracted* by fractional dimension".

*wangxiaonannice@gmail.com
†**Disclaimer:** This summarization is for research and study purposes. It represents a personal interpretation and may contain inaccuracies. Feedback or corrections via email are highly appreciated.

- Reward Scaling;
- Avoiding negative rewards: As negative signals can impede convergence of some RL, division is sign-preserving, compressing the reward magnitude without altering its polarity.

- **Clipped** Fractional Dimension (in post processing):
  - Force to ≥ 1: Preventing agents learning to *fall over immediately* to "game the system";
  - Force to $\leq halfoftopologicaldimension$: Providing a numerically smooth "ceiling" for training to prevent gradients from *spiraling out of control* (*exploding*).

(4) Why is it better than prev.? (Advantages)

- **Low Fractional Dimension** and **More Stable** Trajectories
- Hash Table for Data Compression when Box Meshing
- **Computational Efficient** (Hash Table Box Meshing + Model-Free RL + Fractal Penalty)
- Pursuing Low-Dimensional Trajectories for **Formal Verification**

(5) What is the approach itself?

- Approach Itself:
  - Box Meshing
    - Hash(Mesh Points);
    - Using Hash Table to stire keys.
  - Fractional Dimension Calculation
    - Building $Hsize(statespace) -> 1$ (basically a **decreasing set**)
    - Calculating the Upper & Lower Mesh (Fractal) Dim via the slope of log-log curve (or variation)
  - Post Processing
    - Clipped Mesh Dim
    - Reward **Penalty**: Reward <- Reward / Clip-Dim
- Novelties:
  - **Geometric Regularization**: It introduces fractal dimension as a differentiable reward penalty to steer the agent towards low-dimensional manifolds.
  - **Computational Efficiency**: It proposes a hash-based online box-counting algorithm that enables real-time complexity estimation.
  - **Provable Simplicity**: By actively breaking the "curse of dimensionality," the method produces policies that are not only high-performing but also amenable to formal verification.

(6) What are the applications of it? Legged Lomotion on robotic environments (*HalfCheetah-v2*, *Hopper-v2*, and *Walker2d-v2*) with high dimensionality (11-17 DOF). (Simulation only, real-world robot experiments needed.)

2 The Structure

Summarized Block-Diagram The summarized block-diagram of *Meshing Box RL* see fig. 1[1].

[1] For efficiency, this diagram was initially hand-drawn on paper and then converted into its current digital version using Nano Banana Pro.

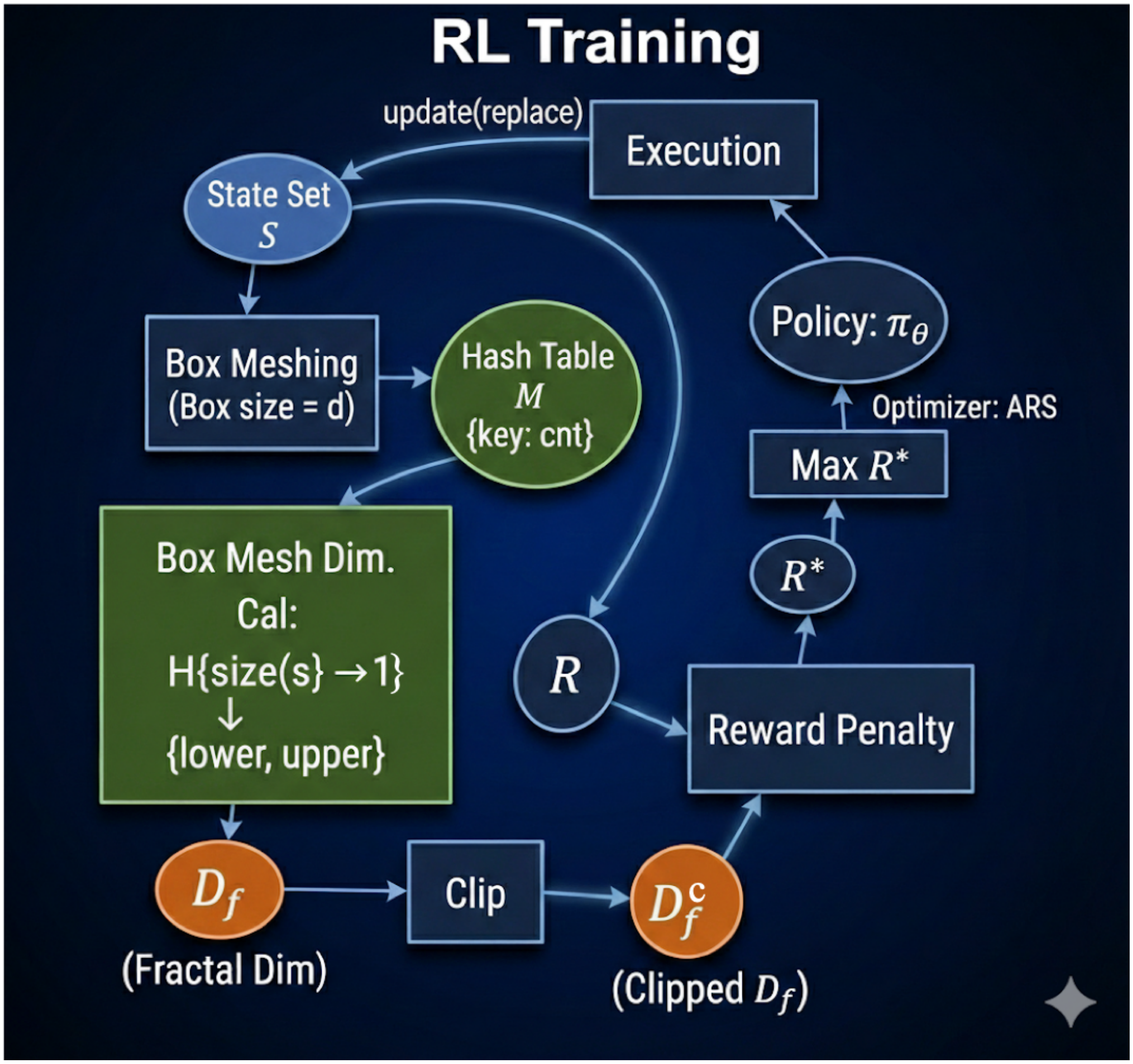

Figure 1: Summarized Block-Diagram of Meshing Box Reinforcement Learning

3 Open Questions

1. **Generalizing to Highly Stochastic Environments:** In highly stochastic environments, such as rocky terrains or slippery surfaces, complex maneuvers are often essential for survival. Does forcibly compressing the fractal dimension risk compromising the agent's performance in high-difficulty settings? Furthermore, is there an "optimal dimension" that strikes a balance between guaranteed verifiability and environmental adaptability?

2. **Beyond the Model-Free RL:** Beyond the field of Model-Free RL, does the concept of Fractal Regularization still hold significant value in legged locomotion based on recent popular approaches?

References

S. Gillen and K. Byl. Explicitly encouraging low fractional dimensional trajectories via reinforcement learning. In *Conference on Robot Learning*, pages 2137–2147. PMLR, 2021.

A Formula Derivation Draft[2]

The fig. 2 shows the "hand-crafted" formula derivation (draft) of Eq 1. when considering the special circumstance (i.e. considering topological dim.).

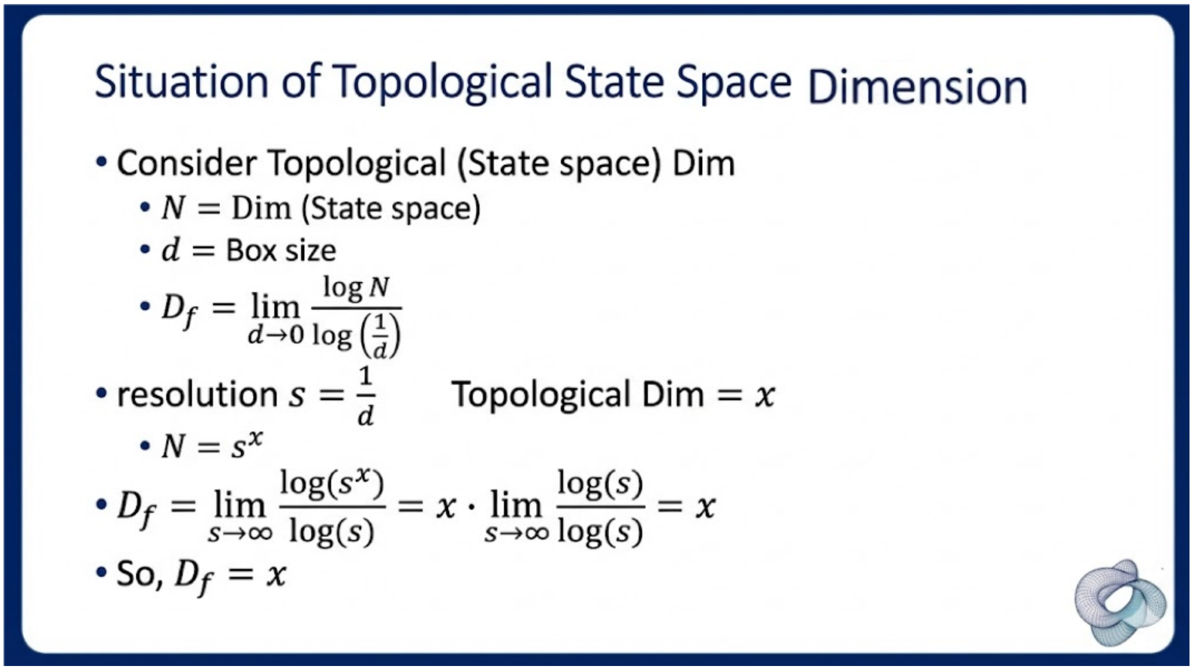

Figure 2: Formula Derivation: When considering about topological dim., Eq. 1 equals to topological dim.

B Note: Physical DOF vs. Feature Dimension vs. Fractal Dimension

The fig. 3[2] shows the "hand-crafted *Mapping the Fea-Dimensionality to Multi-Dimensionality Coordinate Sys.*"

The table 1 illustrates the comparison of "*Physical DOF vs. Feature Dim. vs. Fractal Dim.*"

Table 1: Comparison of Dimensionality Concepts in Robot Locomotion

Concept	Explanation	Metaphor	Attribute
Physical DOF	Degrees of Freedom	The number of **joints** in the robot (hardware constraint).	Constant
Feature Dimension	State Space Dimension	**Total number of coordinate axes** describing the motion (typically $\geq 2\times$ DOF: e.g. for each joint, $\langle q, \dot{q} \rangle$ whereas q is position and $\dot{q}$ is velocity).	Constant
Fractal Dimension	Fractal Dimension (D_f)	The **"thickness" or complexity** of the actual trajectory the robot follows.	**Variable**[3]

[2] Converted by Nano Banana Pro.

[3] Determined by RL policy, *theoretical maximum is State Space Dimension*.

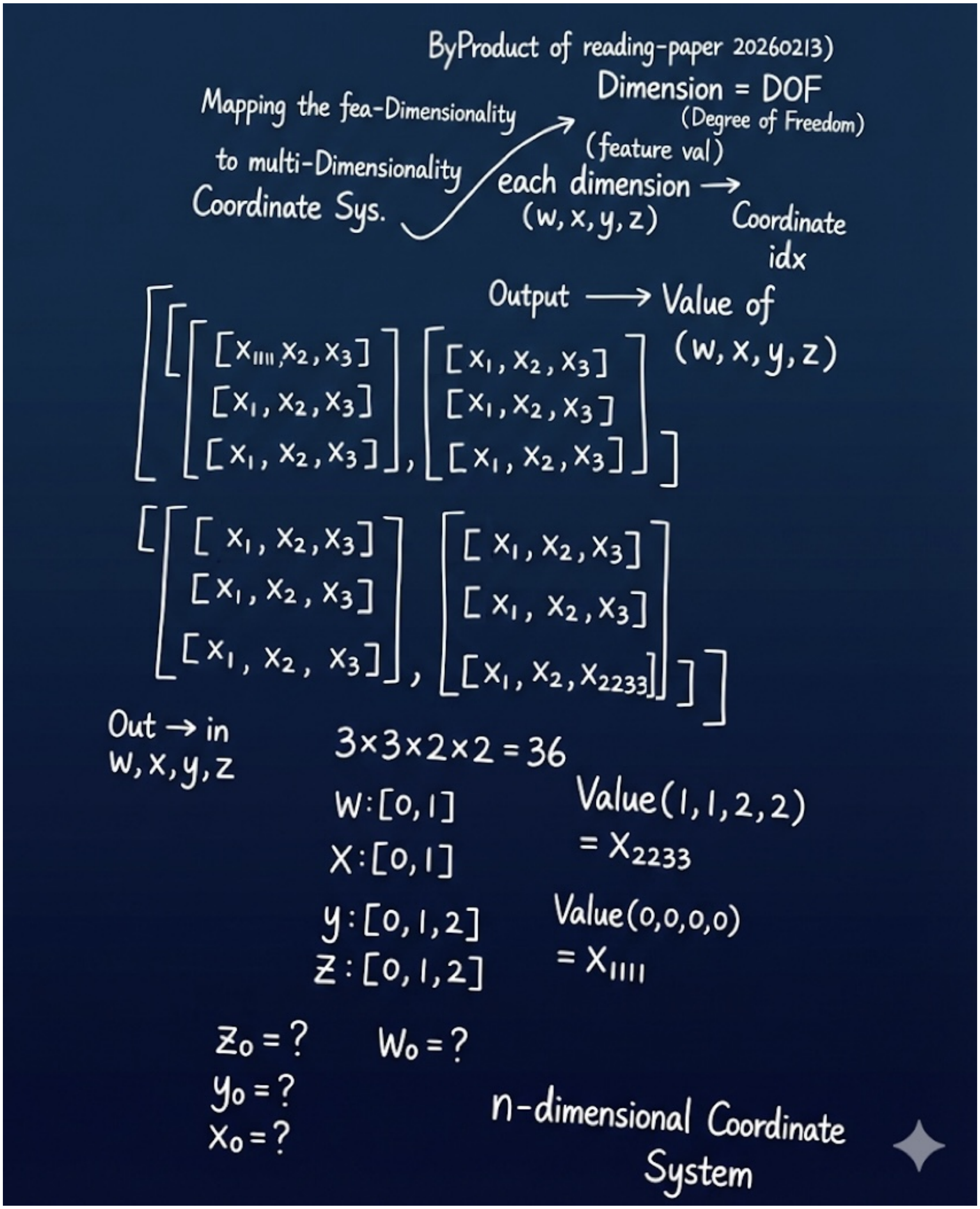

Figure 3: Note Draft: Mapping the Fea-Dimensionality to Multi-Dimensionality Coordinate Sys.